

Novelty Detection in Power Quality Signals with Surrogates: a Time-Frequency Technique

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Abstract - In the new context of smart grids (SGs), new types of electric disturbances may occur in the electrical power systems (EPSs) current and voltage waveforms. The detection and the study of such disturbances can be useful in the operation and control of the EPSs. In order to identify any waveform abnormality or variation, novelty detectors might be used. This work presents a novelty detection technique for power quality signals based on surrogates and distances calculated in the time-frequency domain. Surrogates are signals created with the same magnitude of the Fourier spectrum from the original signal, but with the phase spectrum generated from a random uniform distribution. Tests using several cases of the most common power quality disturbances using synthetic signals were conducted. A comparison with an energy-based novelty detector currently present in the literature is performed, showing better performance results in terms of the receiver operating characteristic (ROC) curve parameters. Also, a real-life waveform signal is used to show the presented technique in action. The surrogate technique presented in this work can be useful in the definition of the novelties ground-truth for real power signal waveforms.

Keywords – surrogate, signal processing, novelty detection, power quality

I. INTRODUCTION

Electric power systems (EPSs) have undergone a significant expansion in recent years [1]. This led to the need for new types of measurement, control, and data transmission instruments as a means of quality and stability maintenance. The modernization and interconnection of the grid through the insertion of new technological equipment is part of the establishment of smart grids (SGs) [2].

SGs allow new possibilities for a more flexible, sophisticated and versatile system, where new types of time-varying nonlinear loads and distributed power sources are present [3]. In this scenario, voltage and current signals may suffer new electrical disturbances, generating new power quality problems, for which EPSs must be prepared.

An important task is to identify any type of change or variation in the voltage and current waveforms, not just the disturbances already defined in the standards and literature (sags, swell, harmonics, transients, etc.) [4, 5], but any new event that may occur in the new context of SGs [6]. Such a task has been termed as "novelty detection". The concept of novelty detection in power quality signals the way it is approached here is relatively recent in the literature [7].

There are some works approaching the novelty detection topic in the literature. In [8], an experimental evaluation of representative novelty detection methods is conducted. A comparative evaluation of four methods is carried out. The experimental results demonstrate that the k -NN novelty detection method exhibits competitive overall performance to the other methods in terms of the receiver operating characteristic (ROC) area under the curve (AUC) metric.

An automatic detector and classifier of voltage waveforms is proposed in [9]. It includes both the classification of the waveforms in multiple known classes and the detection of new waveforms (novelties) that are not available during the training stage. The classification method is based on the Support Vector Data Description (SVDD). The results obtained on both simulated and real-world data demonstrate the ability of the method to identify novelties and to classify known examples correctly.

The work described in [10] presents an approach for single and multiple power quality disturbance detection and classification using multidimensional analysis, higher-order statistics and a neuro-tree based classifier. It is able to detect deviations in the measured voltage waveform from the nominal one and classify 20 classes of single and multiple disturbances with a global efficiency upper to 97%.

An energy-based novelty detector is presented in [7]. This technique is used for comparison purposes in this paper. More details about this energy-based detector functionality are present in Section IV.

In the present work, a technique that uses surrogates adapted to detect novelties in power quality signals is presented. To do so, the detection of novelties is related to the stationary changes in the voltage and current signals. This technique can be useful in the evaluation of real voltage and current waveform signals in order to find the ground-truth of novelties present in them.

The surrogate technique is a time-frequency hypothesis testing tool, where an observation classification into stationary or non-stationary is pursued [11]. Here, we propose an application of this technique to power quality waveform signals, in order to identify the stationary changes related to novelties.

Simulation tests are conducted to characterize the performance of the technique. A real-life waveform signal is used to show the detection capacity of the presented technique.

The article is divided as follows. Section II shows the

surrogate technique. Section III describes the implementation of the surrogate technique for novelty detection in power quality signals. Section IV briefly describes the energy-based novelty detector under comparison. In Section V, some detection results using both simulated and real-life signals are presented. Finally, Section VI states some conclusions.

II. THE SURROGATE TECHNIQUE

A. Definitions

A discrete-time signal can be represented as a vector $\mathbf{x}(n)$, where the index n represents the sample position within the vector $\mathbf{x}(n)$. An N -length observation segment of this signal is a sequence represented by $\{\mathbf{x}_i(n), n = 1, \dots, N\}$. If I observations from the signal $\mathbf{x}(n)$ with length N are considered, we can denote them by $\{\mathbf{x}_i(n), n = 1, \dots, N; i = 1, \dots, I\}$.

Each observation $\mathbf{x}_i(n)$ can be considered as a realization of a discrete-time stochastic process $X(n, i)$ related to the original signal $\mathbf{x}(n)$. Thus, this stochastic process and its realizations can be defined as:

$$\text{Process: } \{X(n, i) : n, i \in \mathbb{N}\} \quad (1)$$

$$\text{Realizations: } X(n, i) = \mathbf{x}_i(n), n = 1, \dots, N; i = 1, \dots, I.$$

Although the concept of stationary processes is uniquely defined in theory, it is interpreted in different ways in practical terms. While in the statistics field stationarity is defined as the invariance of statistical properties over time, it can also be applied to deterministic signals, whose spectral characteristics are invariant over time. In this work, novelty detection is associated with the change of stationarity within an observation. In other words, the change of the spectral characteristics.

B. Surrogates

The objective of the surrogate technique is to analyze a realization in a given observation scale and classify it as belonging to a stationary or non-stationary process, considering it as the only realization of the process.

Surrogates can be understood as a stationary reference for a realization $\mathbf{x}_i(n)$. For the same marginal Fourier spectrum, non-stationary processes are different from stationary processes by a structured organization over time. A set of J surrogates can be computed for an $\mathbf{x}_i(n)$ realization in such a way that each surrogate has the same Power Spectral Density (PSD) as the original signal, while being stationary. In practice, this can be achieved by destroying the organized phase structure that controls non-stationarity, if any.

To create a surrogate $\mathbf{s}_j(n)$, the realization $\mathbf{x}_i(n)$ is first Fourier-transformed. The magnitude of the Fourier spectrum is kept unchanged, while the phase is replaced by a random sequence. On this modified spectrum, the Inverse Fourier Transform is applied, from which the surrogate $\mathbf{s}_j(n)$ is obtained. More surrogates can be created, as many as desired, just performing new random phase replacements and applying the Inverse Fourier Transform, resulting in a set with J surrogates. Creating surrogates as described above forms a stationary reference set for the given realization $\mathbf{x}_i(n)$ [11].

The next paragraph shows the math equations related to the described procedure.

Consider that the realization $\mathbf{x}_i(n)$ has a Discrete Fourier Transform (DFT) $X_i(k)$ with bins indexed by k with a magnitude $A_i(k)$ and phase $\Phi_i(k)$ such that

$$X_i(k) = A_i(k)e^{j\Phi_i(k)} = A_i(k)\angle\Phi_i(k) \quad (2)$$

$$\mathbf{x}_i(n) = \frac{1}{N} \sum_k X_i(k) e^{j\frac{2\pi nk}{N}} = \mathcal{F}^{-1}(X_i(k)),$$

where $\mathcal{F}^{-1}(\cdot)$ stands for the inverse Fourier transform. A surrogate $\mathbf{s}_j(n)$ is created from $X_i(k)$ replacing $\Phi_i(k)$ with an independent, identically distributed (i.i.d.) phase sequence drawn from a uniform distribution over $[-\pi, \pi]$, named $\Psi_j(k)$:

$$\mathbf{s}_j(n) = \frac{1}{N} \sum_k A_i(k) e^{j\Psi_j(k)} e^{j\frac{2\pi nk}{N}} = \mathcal{F}^{-1}(A_i(k) e^{j\Psi_j(k)}). \quad (3)$$

C. Hermite windows

Hermite windows are an important element in the stationarity estimation technique for novelty detection. They are used for windowing the frames and the surrogates. The Hermite windows are a family of functions $h_q(t)$, where q is the window order. The q^{th} -order Hermite window can be calculated as follows:

$$h_q(t) = \left((t - \mathcal{D})^q g(t) \right) / \sqrt{\pi^{\frac{1}{2}} 2^q q!}, \quad (4)$$

where operator $\mathcal{D}$ is the derivative operator $\frac{d(\cdot)}{dt}$ and $g = e^{-\frac{t^2}{2}}$. Fig. 7 shows three Hermite windows, of order 1, 2 and 3.

In practice, a set of windows limited to the Q^{th} -order is used. The choice of Hermite windows is because they are orthonormal and maximally concentrated in the time-frequency domain [12].

D. Hermite spectrograms

A Hermite spectrogram $S_{x_i}^{(h_q)}(t, f)$ can be seen as a convolution of the frame signal $x_i(t)$ with a window formed by the product between the Hermite window $h_q(t)$ and the Fourier complex basis [11]. In other words, it can be considered a short-time Fourier transform (STFT) spectrogram with a Hermite window:

$$S_{x_i}^{(h_q)}(t, f) = \left| \int x(\tau) h_q(\tau - t) e^{-j2\pi f\tau} d\tau \right|^2. \quad (5)$$

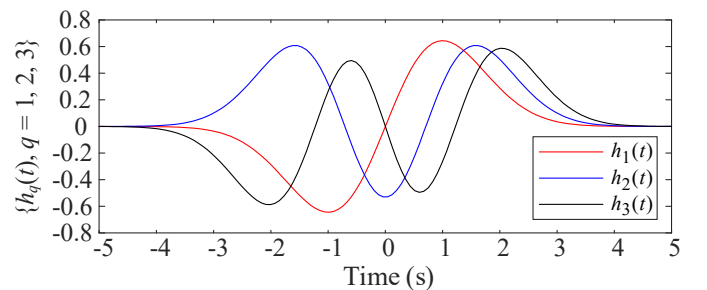


Figure 7. Hermite windows $h_1(t)$, $h_2(t)$ and $h_3(t)$.

In practice, Hermite spectrograms are not calculated in all samples of the frame, meaning that the time resolution is not the sample period T . Instead, it is calculated at D instants of time, each instant represented by t_d , with $d = 1, \dots, D$. The interval of calculation ($t_{d+1} - t_d$) is usually half of the time width of Hermite window. The frequency is discretized taking the discrete values of f_d . Thus, the discrete-time Hermite spectrograms are calculated by replacing the integral in (5) by a summation for each window $\mathbf{h}_q(n)$, as in (6), where $\mathbf{h}_q(n)$ is the discrete-time Hermite window version.

$$S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d) = \left| \sum_{n=1}^N \mathbf{x}_i(n) \mathbf{h}_q(n - \tau) e^{-\frac{j2\pi f_d n}{N}} \right|^2, \quad (6)$$

$$\tau \in \{t_d, d = 1, \dots, D\}, q = 1, \dots, Q,$$

where $\mathbf{h}_q(n - \tau)$ corresponds to the q^{th} -order discrete-time Hermite window shifted by τ samples, and $e^{-\frac{j2\pi f_d n}{N}}$ are the Fourier basis for the analyzed frame.

After calculating the Q Hermite spectrograms, the result are Q matrices, where each column represents a "local spectrum" calculated at each t_d instant.

E. Multitaper spectrograms

With Q Hermite spectrograms, an average of all Q matrices is calculated, element by element, results in a new average matrix, named multitaper spectrogram:

$$S_{\mathbf{x}_i, Q}(t_d, f_d) = \left\langle S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d) \right\rangle_q := \frac{1}{Q} \sum_{q=1}^Q S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d). \quad (7)$$

The multitaper spectrogram is also a matrix of the same order as the Hermite spectrograms. Its columns represent local (average) spectra. After calculating the multitaper spectrogram, all of its columns are averaged over the all t_d instants, resulting in the global (average) spectrum:

$$\left\langle S_{\mathbf{x}_i, Q}(t_d, f_d) \right\rangle_d := \frac{1}{D} \sum_{d=1}^D S_{\mathbf{x}_i, Q}(t_d, f_d). \quad (8)$$

F. Distance test

The global spectrum $\left\langle S_{\mathbf{x}_i, Q}(t_d, f_d) \right\rangle_d$ is a vector whose elements are mean squared amplitudes for each frequency of the Fourier spectrum. Distance quantities can be calculated to compare each local spectrum $S_{\mathbf{x}_i, Q}(t_d, f_d)$ and the global spectrum. Thus, the comparison between the D local spectra and the global spectrum form a set of distances $\{c_d^{(\mathbf{x}_i)}, d = 1, \dots, D\}$, with D elements calculated along all frequencies f_d , defined as:

$$\left\{ c_d^{(\mathbf{x}_i)} := L\left(S_{\mathbf{x}_i, Q}(t_d, f_d), \left\langle S_{\mathbf{x}_i, Q}(t_d, f_d) \right\rangle_d\right), d = 1, \dots, D \right\}, \quad (9)$$

where the distance $L(\cdot, \cdot)$ is composed of two distances, the Kullback-Leibler divergence and the logs-spectral deviation [13].

The calculations of this set of distances is based on the fact

that if there is any variation in the spectrum of $\mathbf{x}_i(n)$ (non-stationarity), the variance of these D distances tends to be a greater value than when $\mathbf{x}_i(n)$ is stationarity.

After calculating the D distances, for the frame $\mathbf{x}_i(n)$, we can calculate the variance of the set $\{c_d^{(\mathbf{x}_i)}, d = 1, \dots, D\}$ as follows:

$$\Theta_{\mathbf{x}_i} = \text{var}\left(c_d^{(\mathbf{x}_i)}\right)_{d=1, \dots, D}, \quad (10)$$

where $\text{var}(\cdot)$ is the variance operator.

G. Non-stationarity detection

After the set of J surrogates is created as described in (3), the same steps described from Subsections D to F are performed for each surrogate $\mathbf{s}_j(n)$. Therefore, this results in J multitaper spectrograms, J global spectra and $J \times D$ local spectra. For each surrogate $\mathbf{s}_j(n)$, a set of distances $\{c_d^{(\mathbf{s}_j)}, d = 1, \dots, D\}$ is calculated as shown in (9), totalizing J distance sets with D distances in each set. As in (10), the variance is calculated for each set, resulting in a set of J variances:

$$\left\{ \Theta_{\mathbf{s}_j} = \text{var}\left(c_d^{(\mathbf{s}_j)}\right)_{d=1, \dots, D}, j = 1, \dots, J \right\} \quad (11)$$

The distribution of the set $\{\Theta_{\mathbf{s}_j}, j = 1, \dots, J\}$ allows the determination of a threshold γ above which the null hypothesis of stationarity is rejected (the observation is non-stationary). The median value of $\{\Theta_{\mathbf{s}_j}, j = 1, \dots, J\}$ can be used as the threshold γ , which is applied to $\Theta_{\mathbf{x}_i}$ taking the form of the following hypothesis test:

$$d(\mathbf{x}_i) = \begin{cases} 1, & \text{if } \Theta_{\mathbf{x}_i} \geq \gamma : \text{"non-stationarity"}; \\ 0, & \text{if } \Theta_{\mathbf{x}_i} < \gamma : \text{"stationarity"} \end{cases}, \quad (12)$$

where $d(\mathbf{x}_i)$ is the decision function for the frame $\mathbf{x}_i(n)$. If $d(\mathbf{x}_i) = 1$ one can assume that a novelty was detected, otherwise, no novelty was detected.

III. SURROGATE TECHNIQUE APPLIED TO NOVELTY DETECTION IN POWER QUALITY SIGNALS

In order to apply the surrogate technique to detect novelties in power quality signals, it is necessary to define the length of the frames $\mathbf{x}_i(n)$ to segment the signal $\mathbf{x}(n)$. The same length used in [7] was adopted, that is, 4 cycles of fundamental frequency. A sampling rate of 128 samples per cycle was used. So, the length of each frame is $N = 512$ samples. Considering the EPS fundamental frequency equal to 60 Hz, the resultant sampling rate is $F_s = 7.68$ kHz and the sampling period is approximately $T = 130 \mu\text{s}$.

A Hermite window width of one fundamental cycle was adopted, and the spectrogram time resolution ($t_{d+1} - t_d$) was half the width of the Hermite window, that is, half a cycle. Thus, the Hermite spectrum is calculated in 7 instants of time within a frame ($t_1, t_2, \dots, t_7$). The maximum Hermite window order of $Q = 4$ was adopted. All implementation and test simulations for this paper were performed in MATLAB[®] environment.

